

Sticky Cantor sets

OPG-37293

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Conjecture Let C be a Cantor set embedded in $\mathbb{R}^n$. Is there a self-homeomorphism f of $\mathbb{R}^n$ for every ϵ greater than 0 so that f moves every point by less than ϵ and $f(C)$ does not intersect C ? Such an embedded Cantor set for which no such f exists (for some ϵ) is called "sticky". For what dimensions n do sticky Cantor sets exist?

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